Hyperspectral Image Cube and Spectral Libraries

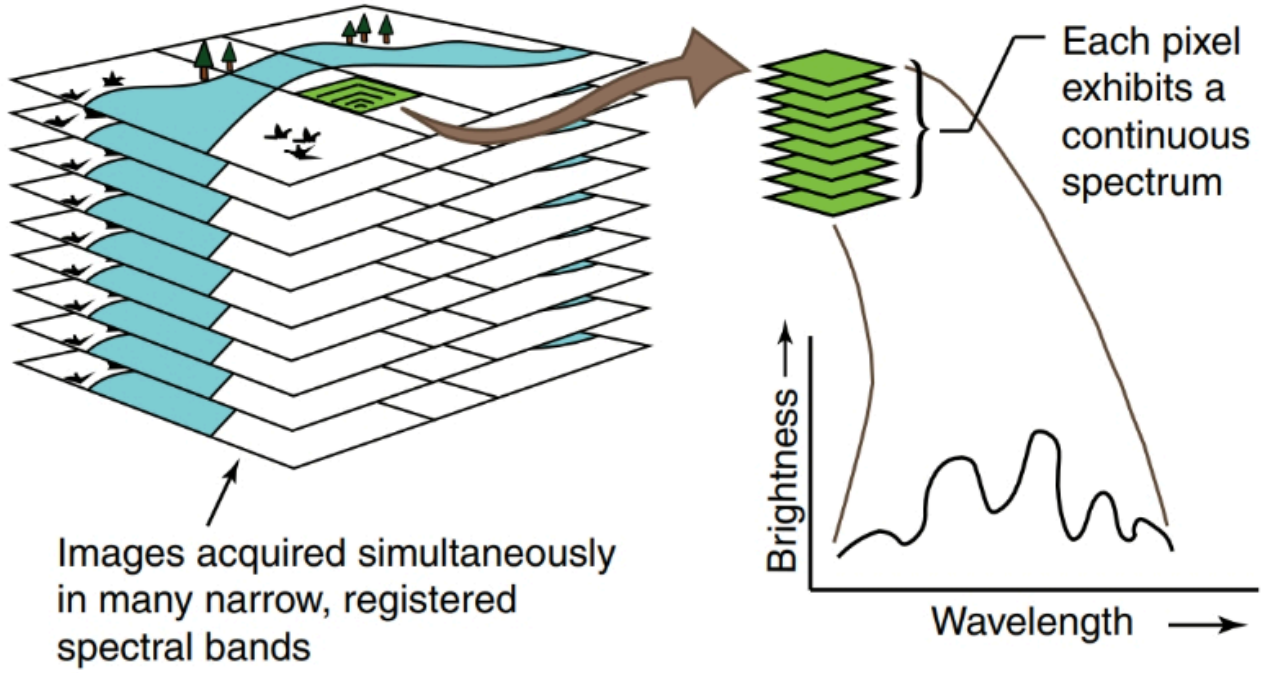
IRS Chapter 14

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CSE620B

Quiz references: **Image Cube, Spectral Libraries** (Week 10).

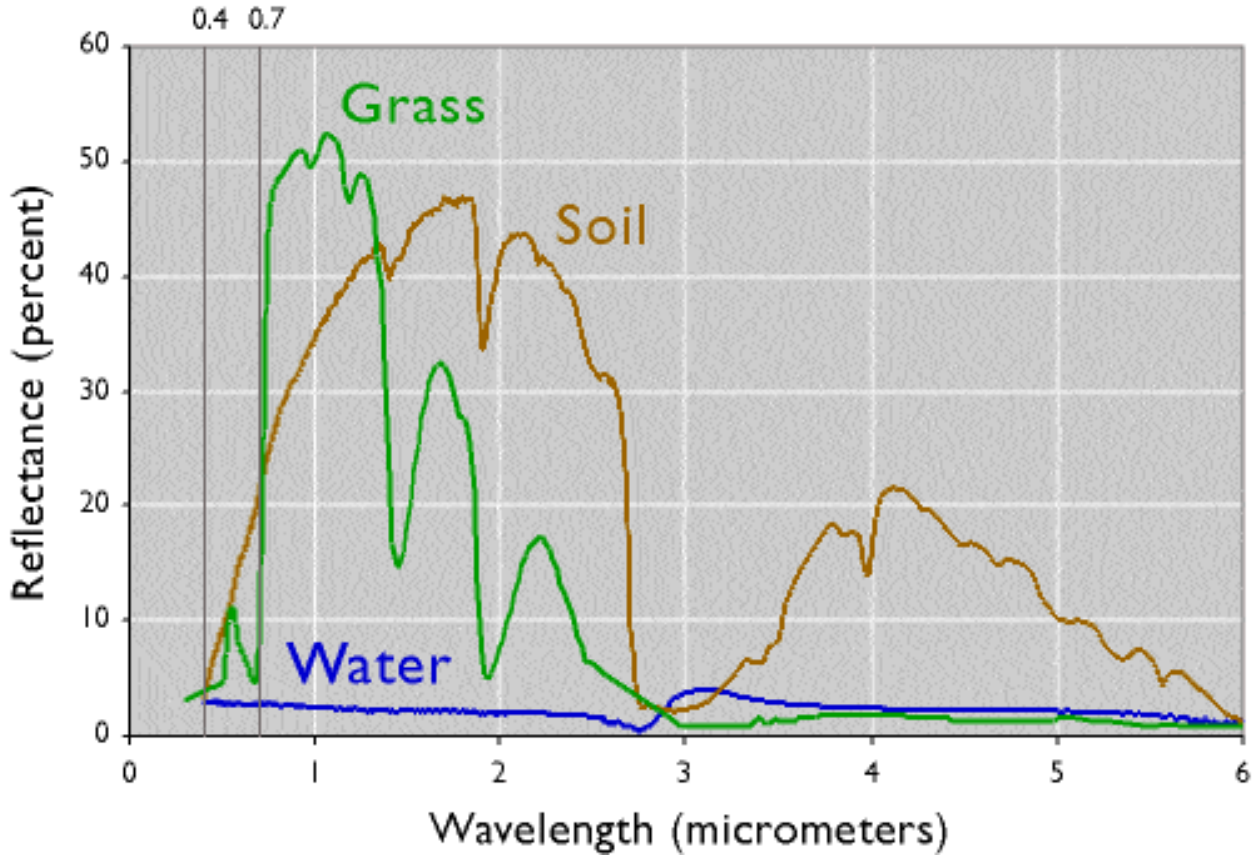
The Hyperspectral Image Cube

- **Image cube**: 3D structure in hyperspectral data
- **X, Y**: Spatial dimensions (map)
- **Z**: Stacked spectral bands (wavelengths)
- Captures **spectral profile for each pixel**



Understanding Spectral Libraries

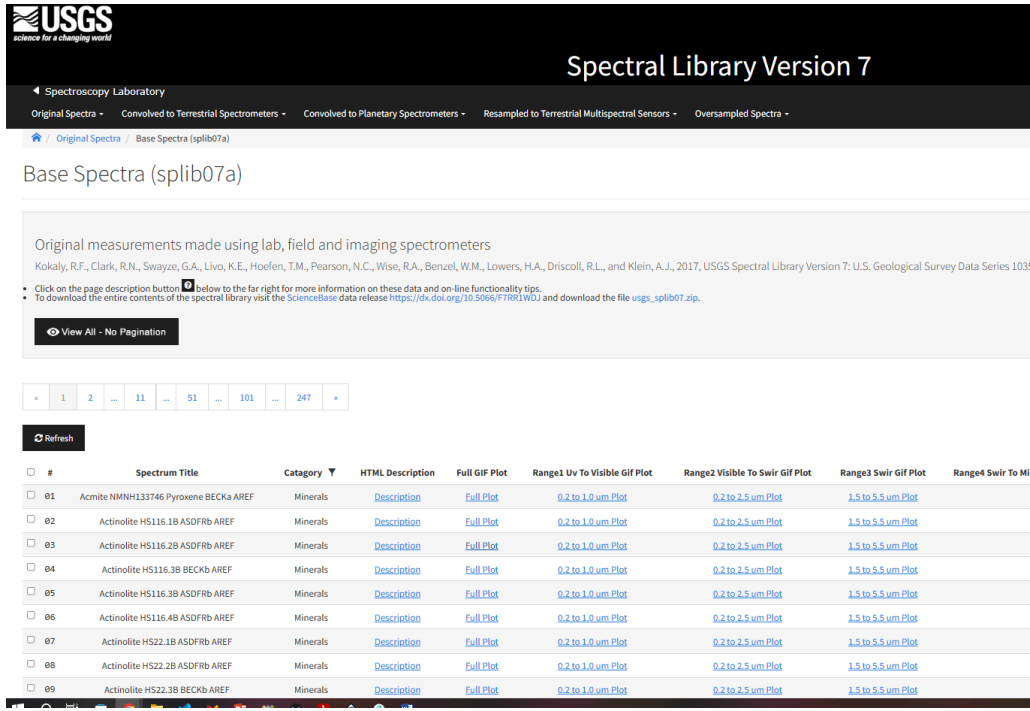
- **Spectral libraries:** Reference databases for spectral analysis
- Store spectra from **lab and field conditions**
- Data from **minerals, vegetation, materials**, etc.
- Essential for **material identification**



Data Sources for Spectral Libraries

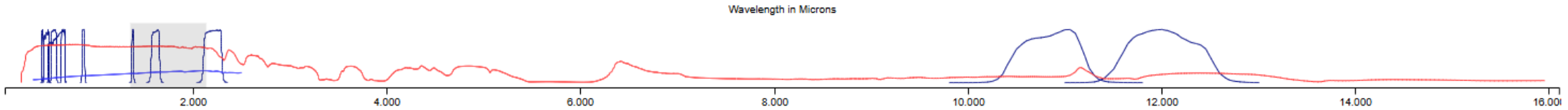
- Data from **laboratory** and **field measurements**
- Includes **minerals, vegetation, materials**, etc.
- Acquired under **controlled and natural conditions**
- Represent **varied terrain and climates**

> Libraries compile spectra from diverse sources to ensure broad applicability in hyperspectral analysis.



Importance of Standardization in Spectral Libraries

- **Standardization** enables consistent use across applications
- Spectra must be comparable despite **instrument and condition differences**
- Data includes **instrument details, conditions, surface characteristics**



> This is the LANDSAT-9 sensor (blue) bands for asphalt(red) spectra.
> https://landsat.usgs.gov/spectral-characteristics-viewer

Abundance Mapping: Key Processing Steps

1. **Convert to surface reflectance**

   (Corrects for atmospheric effects)

2. **Reduce data**: retain the dimensions that carry useful signal

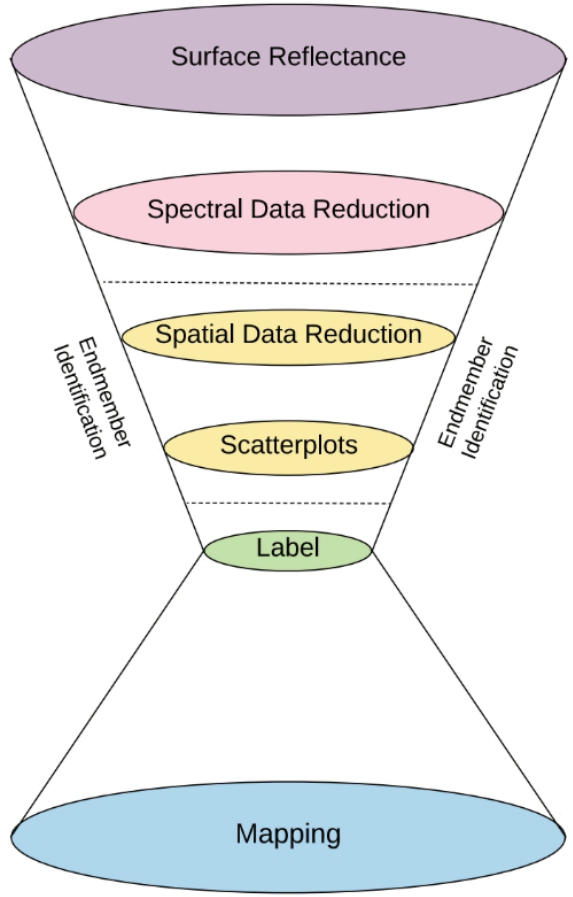
   (MNF orders transformed dimensions using estimated noise; inspect the eigenvalue/noise profile before choosing how many to retain)

3. **Find representative pixels**: Identify "pure" examples

   (These are distinct pixels that define key material signatures)

4. **Estimate material proportions per pixel**

   (Use "pure" pixels to map mixtures across the scene)

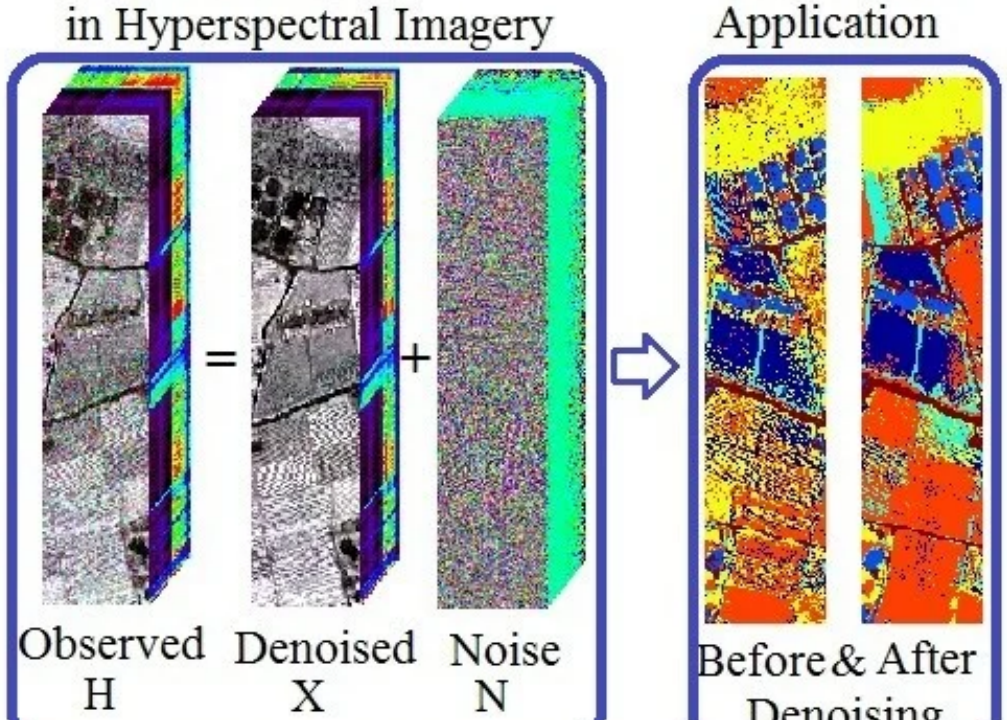


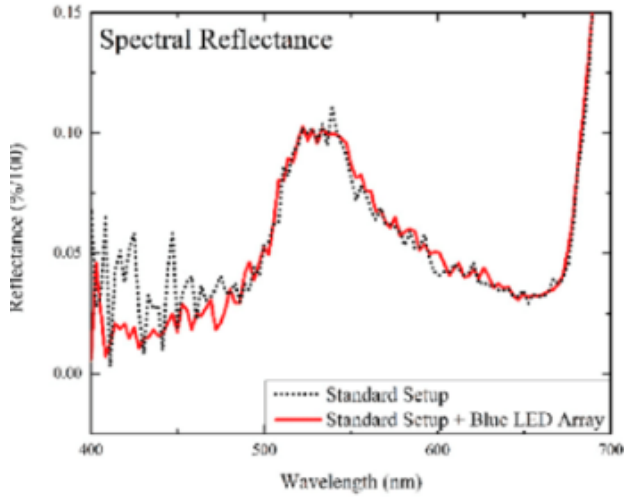
> ℹ️ These steps will be explained in detail in the next slides.

Maximum Noise Fraction (MNF) in Hyperspectral Data

- **Purpose**: Enhance signal by removing noise
- **Key Steps**:
  i. **Estimate noise statistics**: From sensor or homogeneous regions
  ii. **Whiten estimated noise**: Transform its covariance to the identity; this does not remove noise
  iii. **Apply PCA**: Order transformed components by signal-to-noise
- Orders transformed components by signal-to-noise; analysts choose how many to retain
- Essential for improving **material mapping quality**

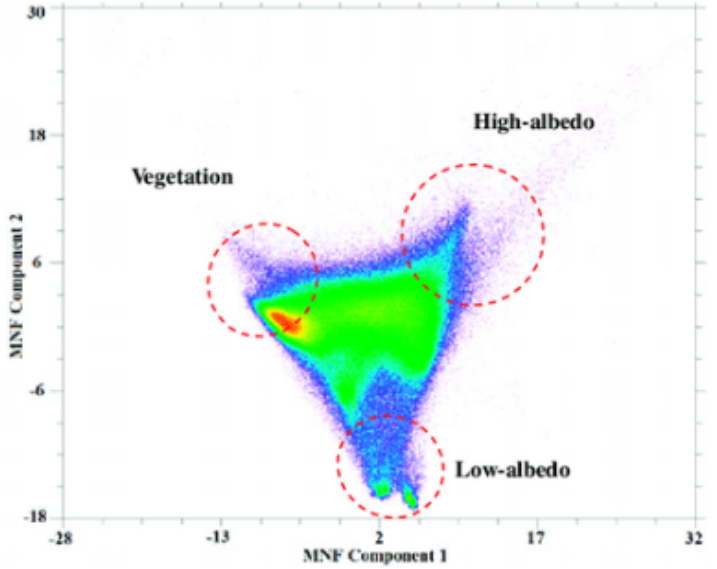
> ℹ️ MNF is a noise-aware extension of PCA.

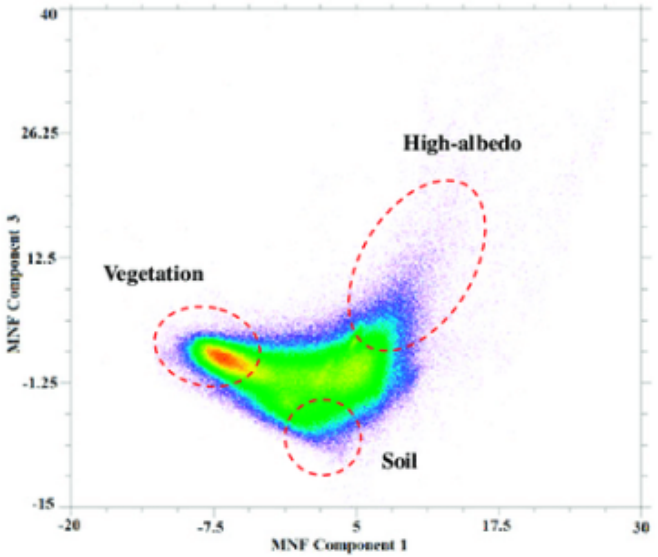




Pure Pixels and Spectral Endmembers

- **Scatter Plot** with each sample an individual pixel (projected)
- **Endmembers**: spectra representing the materials used to explain scene pixels
- Pure pixels can be useful candidates, but they may not exist at the sensor's spatial resolution
- Endmembers can also be estimated with geometric or statistical methods
- Use libraries and field knowledge to interpret candidates; geometry alone does not name a material





Linear Spectral (Un)mixing

- **Purpose**: Estimate material proportions in mixed pixels
- **Fraction vector** (f): Contains proportions of each material
- **Constraints**: Fractions are *nonnegative* and *sum to 1* when using the fully constrained model

$$\mathbf{x} = \sum_{j=1}^{N} f_j \mathbf{m_j} = [\mathbf{m_1}, \mathbf{m_2}, \ldots, \mathbf{m_N}] \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{bmatrix} = \mathbf{Mf}$$

$$\text{s.t.} \quad f_j \geq 0, \qquad \sum_j f_j = 1$$

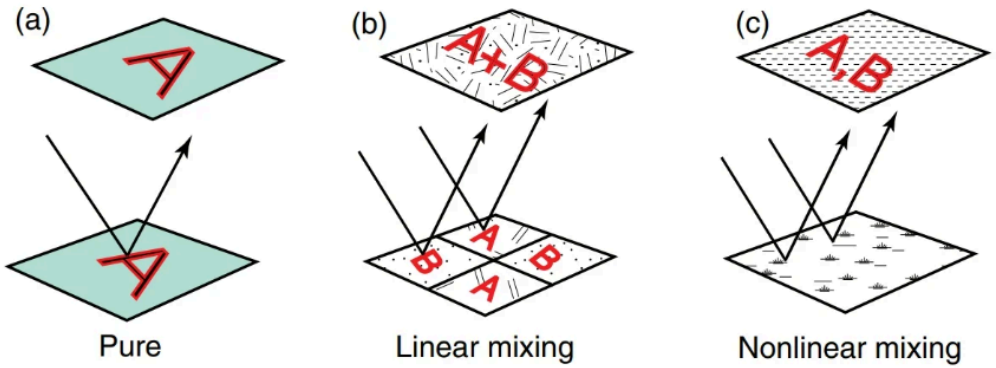


TABLE 14.1 Example of Mixture of Two Endmembers

	Water	Trees	Riparian
Blue	1014.0	941.0	962.9
Red	832.0	799.0	808.9
Green	538.0	579.0	566.7
NIR	338.0	1846.0	1393.6

Quiz Questions 15, 16